# MUHAMAD HENDRI FEBRIANSYAH

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## PROFESSIONAL SUMMARY

Senior Software Engineer with 6+ years of experience building high-performance web, mobile, and backend systems, with practical background in AI/ML workflows. Proven track record building real-time mission-critical dashboards for NTMC Korlantas Polri (< 500ms streaming latency), optimizing native Android applications for 1.5M+ active users (38.6% size reduction, 99.42% crash-free rate), and integrating AI-powered features including conversational agents and knowledge base pipelines. Proficient in Python, SQL, Docker, Linux, and Git, with certifications in Machine Learning and Data Analysis from Dicoding Indonesia.

## WORK EXPERIENCE

**Full Stack Developer (Web3 & AI) | Kipley Pte. Ltd. (Singapore (Remote))** *Nov 2024 – Jul 2025*

- Engineered and orchestrated high-throughput **REST & GraphQL APIs** across 5+ flagship dApps, boosting frontend response speeds by **30%** through optimized caching layers.
- Architected seamless **Ethereum Smart Contract (Solidity)** integrations with **Next.js 15 (App Router, Turbopack)** frontends via **Wagmi v2** and **Ethers.js v6** for secure DeFi/NFT transactions.
- Developed an autonomous **Telegram AI Assistant** supporting multi-chain settlement (**TON, USDT, Telegram Stars**) and fiat on-ramp integrations with **100% zero-failure transaction reliability**.
- Championed Agile sprint execution, collaborating cross-functionally with UI/UX and product teams to deliver feature releases on schedule.

**Freelance Software Engineer | PT Lapantiga Solusi Algoritma (Surabaya, Indonesia)** *Jan 2019 – Present*

- Architected and deployed the **NTMC Command Center Dashboard Utama**, an operational traffic & incident monitoring platform utilizing **React 19**, **React Router v7**, **TypeScript**, **Tailwind CSS v4**, and **shadcn/ui**.
- Spearheaded real-time bidirectional streaming via **Pusher WebSockets**, achieving **< 500ms incident broadcast latency** and instant multi-screen telemetry synchronization.
- Engineered high-performance server-side data tables with **TanStack Table v8** and server-state caching via **TanStack Query v5**, supporting complex multi-field filtering across large datasets.
- Designed 100% type-safe form workflows using **React Hook Form** with strict **Zod** schema validation and interactive visualization charts via **Recharts**.
- Delivered robust enterprise backends, virtual exhibition platforms (**Efosh**), tax e-Bupot systems, and **Petrokimia Gresik** insurance claim portals leveraging **Laravel 12/10**, **Livewire**, **AWS S3**, and **MySQL**.

**Software Developer | PT Aku Pintar Indonesia (Surabaya, Indonesia)** *Feb 2021 – May 2024*

- Spearheaded native **Android (Kotlin & Java)** development for a flagship EdTech platform serving **1,500,000+ active users** with a **4.42/5.00 Google Play Store rating**.
- Architected a modular **Clean Architecture & MVVM** refactor using **Android Jetpack**, shrinking application size by **38.6%** (**69.5 MB to 42.7 MB**) and boosting runtime rendering performance.
- Optimized core background sync and memory allocation to sustain a **99.42%+ crash-free session rate** across continuous 90-day production release cycles.
- Maintained enterprise **Liferay OSGi web portal modules** (100+ micro-modules), integrating national **RIASEC psychometric testing**, **WebRTC live video counseling**, and automated payment gateways (**BCA, OVO**).

**Software Developer | PT Qira Teknologi Indonesia (Surabaya, Indonesia)** *Nov 2020 – Jan 2024*

- Architected scalable web backends for **Kemdikbudristek RI (EPAK Widyaprada)** national civil servant credit scoring, automotive e-commerce, and CRM platforms using **PHP Laravel**, **MySQL**, and **AWS S3**.
- Engineered secure payment gateway integrations (**Midtrans Snap**, **Moota**) and Google APIs, guaranteeing **99.9% billing transaction accuracy** and automated reconciliation.
- Developed companion native Android applications utilizing **Kotlin**, **Android Jetpack**, and **Firebase** for real-time data synchronization.
- Configured CI/CD deployment pipelines on **Linux/SSH**, **cPanel**, and **Plesk**, ensuring high availability and zero-downtime production rollouts.

## TECHNICAL SKILLS

**Computer Vision & OCR:** OpenCV, YOLO (v8/v11), PaddleOCR, Tesseract OCR, TensorFlow / Keras (CNN), Image Preprocessing, Object Detection

**AI, ML & Edge Optimization:** Predictive Modeling, Statistical Analysis, Scikit-learn, Model Quantization (TFLite / ONNX), Edge Inference (Android / Mobile Devices), RAG Pipelines

**Data Pipelines & MLOps:** Python (NumPy, Pandas, Matplotlib), SQL (PostgreSQL, MySQL), MLflow, Apache Airflow, Docker, Linux/SSH, Git / GitHub Actions

**APIs & Operational Integration:** RESTful & GraphQL APIs, KoboToolbox API Integration, Live Operational Dashboards (WebSockets, SSE), FastAPI, Flask, Redis

**Mobile & Edge Platform:** Android Jetpack (Kotlin/Java), Room SQLite, Offline-first Caching, Memory Leak Profiling, ProGuard/R8 Optimization

## EDUCATION

**Institut Teknologi Sepuluh Nopember (ITS) | Bachelor of Computer Science (S.Kom.) in Informatics Engineering** *Jul 2014 – Aug 2018*

GPA: 3.42 / 4.00 | Focus Area: Network Centric Computing & Distributed Systems

## CERTIFICATIONS

**Belajar Machine Learning untuk Pemula | Dicoding Indonesia / Google Developers** *Aug 2024 – Aug 2027*

Core Focus: Supervised & Unsupervised Learning, SVM, Neural Networks, CNN Image Classification, TensorFlow

**Belajar Dasar AI | Dicoding Indonesia / Google Cloud Partner** *Aug 2024 – Aug 2027*

Core Focus: Artificial Intelligence Fundamentals, Machine Learning Concepts, Deep Learning Principles

**Belajar Analisis Data dengan Python | Dicoding Indonesia / Google Developers** *Jan 2024 – Jan 2027*

Core Focus: Descriptive Statistics, Data Wrangling, Exploratory Data Analysis (EDA), Data Visualization, Streamlit

**Belajar Dasar Google Cloud | Dicoding Indonesia / Google Cloud Partner** *Aug 2024 – Aug 2027*

Core Focus: Google Cloud Platform (GCP), Compute Engine, Cloud Storage, VPC, Cloud IAM & Security

## LANGUAGES

**Indonesian:** Native proficiency **English:** Professional working proficiency